

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Led **root cause analysis** and **corrective action implementation** for yield and performance issues across enterprise production systems, collaborating with cross-functional defect reduction, device engineering, and yield analysis teams to meet committed production targets within SLA windows.
- Analyzed **structured and unstructured data** using **Python, SQL**, and **statistical methods** to deliver actionable insights into process health, enabling data-driven decisions on continuous improvement activities and yield optimization.
- Established and implemented **quality assurance workflows and procedures** for continuous improvement programs, preventing yield excursions in line and sort processes through process standardization and automated monitoring via **CloudWatch, Splunk**, and **Dynatrace**.
- Collaborated with Technology Development and local yield teams to **import new technology to production environments**, managing product introduction, optimization of yield and performance targets, and change-control coordination across interconnected ERP and enterprise business systems.
- Assessed quality and reliability risks for manufacturing process changes and material updates, ensuring adherence to regulatory standards and compliance frameworks through cross-functional collaboration with operations teams and executive stakeholders.
- Owned product maintenance across multiple production systems, driving **continuous improvement** through data mining, statistical analysis, and reliable problem-solving in high-pressure, fast-paced manufacturing environments with demonstrable cross-functional leadership.

SKILLS

Data Analysis & Yield Engineering – root cause analysis, defect reduction, yield optimization, process health analytics, statistical analysis, SLA management.

Manufacturing & Process Technologies – FEOL/BEOL integration, yield analysis, quality assurance workflows, continuous improvement procedures, process standardization, product maintenance.

Data & Analytics – Python, SQL, pandas, NumPy, scikit-learn, data mining, hypothesis testing, regression analysis, Power BI, Tableau, DAX, Power Query.

Enterprise Systems & Integration – ERP integration support, change-control management, cross-system data flows, REST APIs, PostgreSQL, MySQL, Oracle, system interconnections.

Production Monitoring & Reliability – CloudWatch, Splunk, Dynatrace, incident response, P1/P2 triage, operational alerting, process monitoring.

Development & DevOps – Java Spring Boot, Python scripting, AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD pipelines, Linux.

Communication & Collaboration – cross-functional team leadership, technical documentation, stakeholder presentations, requirements gathering, UAT coordination, regulatory compliance.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Production Yield & Performance Analytics Dashboard (HCL Technologies)

- Designed and deployed a comprehensive **Power BI dashboard** integrating production monitoring data from **Splunk, Dynatrace, and CloudWatch** to visualize yield trends, performance metrics, and process health across enterprise manufacturing systems.
- Developed **DAX measures** and **Power Query ETL workflows** to aggregate structured production data from **PostgreSQL** and **Oracle** databases, enabling real-time root cause analysis and SLA compliance tracking for production yield targets.
- Automated **continuous improvement reporting** through **Python** data pipelines and scheduled SQL queries, reducing manual analysis time by **40%** and enabling faster decision-making for corrective action implementation across operations and yield teams.

Manufacturing Process Integration & Change-Control System (HCL Technologies)

- Built an enterprise integration platform using **Java Spring Boot** REST APIs and **Python automation scripts** to manage product maintenance workflows, technology imports from development to production, and change-control procedures across interconnected ERP and business systems.
- Implemented **data validation and quality assurance procedures** via **SQL schema changes** and automated testing, ensuring manufacturing process changes and material updates adhered to regulatory standards and risk assessment requirements.
- Collaborated with cross-functional defect reduction, device engineering, and operations teams to document and standardize quality workflows, reducing yield excursion incidents by **30%** and improving product introduction timelines through reliable change-management automation.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)